Physical Realization and Predeclared Experimental Test of a Clock-Free HSMT Dynamical Benchmark

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Abstract

We present the experimental protocol for a physical realization of a minimal conditional dynamical benchmark developed within Hierarchical Shell Manifold Theory (HSMT).

The benchmark is based on the three-state path graph P_3, with structural Laplacian

$$K = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix},$$

whose positive eigenvalues are 1 and 3.

Two conditional dynamical realizations constructed from the same structural operator make distinct clock-independent predictions,

$$\rho_{\mathrm{S}} = 3, \qquad \rho_{\mathrm{I}} = \sqrt{3}.$$

The experiment realizes both models in the same six-state analog apparatus. The structural networks, integrator banks, preparation, modal readouts, hysteretic event detectors, and observation protocol are shared between the two runs. Only the upper dynamical feedback branch is changed.

To prevent circular inference, the experiment is divided into a structural calibration phase and a subsequent blinded dynamical phase. The physical matrices K_A, K_B, H_B, and the two integrator gain matrices G_u and G_v are measured before the transient records used for model discrimination are examined. These measurements determine the calibrated state matrices and the pre-run prediction bands.

The formal observable is an internally synchronized event-count ratio. The observation window begins on a completed reference-channel event and spans a predeclared integer number M of complete reference periods. No calibrated external time standard is needed for the primary classification.

This manuscript fixes the apparatus, calibration procedure, observation rule, acceptance criteria, failure criteria, and reporting requirements before physical data are acquired. No experimental result is claimed in the present version.

1 Introduction

A theoretical model becomes experimentally meaningful only when its predictions can be connected to a physical measurement protocol whose rules are fixed before the relevant data are examined.

Previous work developed a minimal conditional benchmark within the HSMT framework. The benchmark begins with a three-state relational configuration graph and produces two distinct dynamical predictions from the same structural operator:

$$\rho_{\mathrm{S}} = 3$$

and

$$\rho_{\mathrm{I}} = \sqrt{3}.$$

Those predictions are dimensionless ratios of modal rates.

The same analysis showed that the two candidate models can be realized in a common six-real-state analog architecture and distinguished using internally generated hysteretic event counts.

The purpose of the present paper is not to derive that benchmark again.

The purpose is to fix the physical experiment that will test it.

The experimental chain is

$$\begin{aligned}\text{physical circuit} &\longrightarrow \text{static calibration} \longrightarrow \text{pre-run prediction} \\ &\longrightarrow \text{physical transient} \longrightarrow \text{event record} \longrightarrow \text{comparison}.\end{aligned}$$

The order of these steps is essential.

In particular, the transient record may not be used to redefine the structural matrices that are subsequently claimed to predict that same record.

The experiment is therefore constructed around a strict calibration–prediction firewall.

2 Scope of the experiment

The experiment tests a narrow statement.

It asks whether a physical implementation of the previously derived conditional benchmark behaves according to the dynamics predicted from independently calibrated circuit structure.

A successful experiment would demonstrate a physical realization of

$$\text{structural operator} \longrightarrow \text{conditional dynamics} \longrightarrow \text{clock-free event record}.$$

It would not establish:

- HSMT as a fundamental theory of nature;
- a physical identification of hierarchy grade N;
- physical $3+1$-dimensional spacetime;
- Gaussian shell profiles;
- cosmological shell structure;
- dark matter or dark energy interpretations;
- a unique fundamental master dynamics.

The experiment is therefore an operational benchmark rather than a fundamental-physics confirmation experiment.

3 Structural benchmark

The physical apparatus is based on the matrix

$$K = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}.$$

Its eigenvalues are

$$\boxed{0, \quad 1, \quad 3.}$$

An orthonormal eigenbasis is

$$\phi_0 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix},$$

$$\phi_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix},$$

and

$$\phi_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}.$$

The zero mode is excluded from the dynamical comparison by the common preparation. The two positive structural modes therefore correspond to

$$\lambda_1 = 1$$

and

$$\lambda_2 = 3.$$

4 Predeclared candidate dynamics

Two dynamical realizations are tested.

4.1 S dynamics

The ideal S realization is

$$\dot{u} = gKv,$$

$$\dot{v} = -gKu.$$

The modal rates are

$$\Omega_j^{(\mathrm{S})} = g\lambda_j.$$

Thus

$$\Omega_1^{(\mathrm{S})} = g, \qquad \Omega_2^{(\mathrm{S})} = 3g,$$

and the primary prediction is

$$\boxed{\rho_{\mathrm{S}} = 3.}$$

4.2 I dynamics

The ideal I realization is

$$\dot{u} = gv,$$

$$\dot{v} = -gKu.$$

Eliminating v gives

$$\ddot{u} + g^2 K u = 0.$$

Therefore

$$\Omega_j^{(\mathrm{I})} = g\sqrt{\lambda_j},$$

and

$$\boxed{\rho_{\mathrm{I}} = \sqrt{3}.}$$

Numerically,

$$\sqrt{3} \approx 1.7320508.$$

The ideal model separation is therefore

$$\boxed{\Delta_{\mathrm{ideal}} = 3 - \sqrt{3} \approx 1.2679492.}$$

5 Six-state physical architecture

The state of the apparatus is

$$Y = \begin{pmatrix} u \\ v \end{pmatrix} \in \mathbb{R}^6,$$

where

$$u = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}, \qquad v = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}.$$

The physical circuit contains two three-channel integrator banks. The lower feedback path implements a physical approximation

$$K_A$$

to the ideal matrix K.
The upper S path implements a second physical approximation

$$K_B.$$

The upper I path implements

$$H_B$$

with ideal target

$$H_B = bI.$$

The measured integrator gain matrices are

$$G_u = \mathrm{diag}(g_{u1}, g_{u2}, g_{u3}),$$

and

$$G_v = \mathrm{diag}(g_{v1}, g_{v2}, g_{v3}).$$

The physical S equations are

$$\boxed{\dot{u} = G_u K_B v, \qquad \dot{v} = -G_v K_A u.}$$

The physical I equations are

$$\boxed{\dot{u} = G_u H_B v, \qquad \dot{v} = -G_v K_A u.}$$

Only the upper feedback selection changes between the two tests.

6 Nominal analog implementation

The nominal six integrators use

$$R_I \approx 100\,\mathrm{k\Omega}$$

and

$$C_I \approx 100\,\mathrm{nF}.$$

The resulting nominal rate scale is

$$g \approx \frac{1}{R_I C_I} \approx 100\,\mathrm{s}^{-1}.$$

The corresponding ideal frequencies are therefore approximately

$$f_1 = \frac{100}{2\pi} \approx 15.92\,\mathrm{Hz},$$

$$f_2^{(\mathrm{S})} = \frac{300}{2\pi} \approx 47.75\,\mathrm{Hz},$$

and

$$f_2^{(\mathrm{I})} = \frac{100\sqrt{3}}{2\pi} \approx 27.57\,\mathrm{Hz}.$$

These frequencies are engineering expectations only.
The primary experimental observable is not an externally measured frequency.

7 Physical implementation of K

For

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix},$$

the matrix action can be written

$$Kx = \begin{pmatrix} x_1 - x_2 \\ -x_1 + 2x_2 - x_3 \\ x_3 - x_2 \end{pmatrix}.$$

Define

$$q_1 = x_1 - x_2,$$

and

$$q_3 = x_3 - x_2.$$

Then

$$q_2 = -(q_1 + q_3).$$

Thus

$$Kx = \begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix}.$$

Each physical K block can therefore be implemented using two difference channels and one inverting summing channel rather than a general matrix multiplier.

8 Common preparation

Both candidate dynamics use the same initial state,

$$\boxed{u(0) \propto \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \qquad v(0) = 0.}$$

Because

$$1 - 1 + 0 = 0,$$

the preparation is orthogonal to the zero mode.
Both positive eigenmodes are excited.
The experiment does not require an exact absolute preparation amplitude.
It requires:

- no significant zero-mode contamination;
- both positive modes to be present;
- sufficient amplitude to cross the hysteretic thresholds;
- no saturation of the analog stages.

9 Common modal readouts

The reference readout is

$$\boxed{X_1 = u_1 - u_3.}$$

The test readout is

$$\boxed{X_2 = \frac{u_1 - 2u_2 + u_3}{3}.}$$

In the ideal system,

$$X_1(t) = \cos(gt)$$

for both candidate dynamics.
The test channel is

$$X_2^{(\mathrm{S})}(t) = \cos(3gt)$$

for S and

$$X_2^{(\mathrm{I})}(t) = \cos(\sqrt{3}\, gt)$$

for I.
The reference channel therefore provides an internal control.
A material S/I change in reference behavior is treated as an experimental anomaly rather than evidence for either candidate model.

10 Hysteretic event detector

Each modal signal is converted into events using two thresholds,

$$+h$$

and

$$-h,$$

with nominal value

$$\boxed{h = 0.5\,\mathrm{V}.}$$

The detector begins in a high state.

A transition to the low state occurs only after

$$X \leq -h.$$

A completed event occurs only after the subsequent crossing

$$X \geq +h.$$

The detector then returns to the high state.

This ordered sequence suppresses small-amplitude threshold chatter and generates one completed event per sufficiently clean oscillation.

11 Mode-purity acceptance condition

Let an experimentally observed modal channel be represented as

$$X(t) = A\cos(\Omega t + \phi) + \eta(t),$$

where $\eta(t)$ includes contamination from unwanted modes, offsets, and noise.

A sufficient predeclared hysteretic acceptance condition is

$$\boxed{\|\eta\|_\infty < 0.5}$$

and

$$\boxed{A > 0.5 + \|\eta\|_\infty.}$$

A channel that fails this condition is not accepted for the formal event-count comparison.

12 Experimental control states

The apparatus uses distinct logical phases.

12.1 RESET

The integration capacitors are discharged.

Both state banks are held.

12.2 PREP

The u bank is prepared along

$$(1, -1, 0)^T.$$

The v bank remains zero.

12.3 RUN

The preparation source is disconnected.

The selected S or I feedback branch is connected.

Both integrator banks evolve freely.

The experiment therefore follows

$$\boxed{\text{RESET} \longrightarrow \text{PREP} \longrightarrow \text{RUN}.}$$

13 Static calibration firewall

The most important methodological requirement is that physical structure must be measured before dynamical prediction data are examined.

The following quantities are therefore calibrated first:

$$\boxed{G_u, \quad G_v, \quad K_A, \quad K_B, \quad H_B.}$$

Only after these quantities have been recorded and locked may the formal S and I transient records be analyzed.

14 Integrator-gain calibration

For each integrator channel i, apply equal positive and negative test voltages.

Let

$$s_i^+$$

be the measured output slope under $+V_T$,

and let

$$s_i^-$$

be the slope under $-V_T$.

The measured gain is

$$\boxed{g_i = \frac{s_i^- - s_i^+}{2V_T}.}$$

The use of equal positive and negative ramps suppresses constant first-order offset contributions.

The six resulting gains form

$$G_u$$

and

$$G_v.$$

15 Matrix calibration

Each structural network is calibrated through bipolar basis-vector injection.

For basis vector e_j, apply

$$+V_T e_j$$

and

$$-V_T e_j.$$

Let the corresponding outputs be

$$y_j^+$$

and

$$y_j^-.$$

Then column j of the measured matrix is

$$\boxed{M_{\cdot j} = \frac{y_j^+ - y_j^-}{2V_T}.}$$

Applying this procedure to each network produces

$$K_A,$$

$$K_B,$$

and

$$H_B.$$

No oscillation-frequency information is used in this calibration.

16 Scalar and nonscalar error decomposition

For K_A, define the best scalar fit to the ideal matrix by

$$a = \frac{\langle K_A, K \rangle_F}{\|K\|_F^2}.$$

Then define

$$R_A = K_A - aK.$$

Similarly,

$$b = \frac{\langle K_B, K \rangle_F}{\|K\|_F^2},$$

and

$$R_B = K_B - bK.$$

The scalar coefficients a and b alter primarily the overall clock scale.
The residual matrices

$$R_A, \qquad R_B$$

represent the spectrally relevant shape distortions.
This decomposition is recorded before the transient run.

17 Calibrated prediction matrices

The physical S state matrix is constructed from the static calibration:

$$\boxed{A_{\mathrm{S}}^{\mathrm{cal}} = \begin{pmatrix} 0 & G_u K_B \\ -G_v K_A & 0 \end{pmatrix}.}$$

The physical I state matrix is

$$\boxed{A_{\mathrm{I}}^{\mathrm{cal}} = \begin{pmatrix} 0 & G_u H_B \\ -G_v K_A & 0 \end{pmatrix}.}$$

Their positive oscillatory eigenvalues determine the pre-run predicted modal rates.
From those rates define

$$\rho_{\mathrm{S}}^{\mathrm{cal}}$$

and

$$\rho_{\mathrm{I}}^{\mathrm{cal}}.$$

These values are written into the experiment record before the classification transient is examined.

18 Predeclared observation length

Let the calibration uncertainty produce ratio uncertainties

$$\delta\rho_{\mathrm{S}}$$

and

$$\delta\rho_{\mathrm{I}}.$$

Define

$$\Delta_{\mathrm{cal}} = \left| \rho_{\mathrm{S}}^{\mathrm{cal}} - \rho_{\mathrm{I}}^{\mathrm{cal}} \right|.$$

The observation length M must satisfy

$$\boxed{\Delta_{\mathrm{cal}} > \frac{2}{M} + \delta\rho_{\mathrm{S}} + \delta\rho_{\mathrm{I}}.}$$

Equivalently,

$$M > \frac{2}{\Delta_{\mathrm{cal}} - \delta\rho_{\mathrm{S}} - \delta\rho_{\mathrm{I}}}.$$

The selected integer M is recorded before the transient data are used for classification.
For the ideal system,

$$M = 2$$

is sufficient.

19 Synchronized event window

Counters are not formally started at the arbitrary release time.
Instead, completed reference events are denoted

$$r_1, r_2, r_3, \ldots.$$

The first completed reference event defines synchronization.
The formal observation interval is

$$\boxed{(r_1, r_{M+1}].}$$

This interval contains exactly M complete reference periods.
If the test channel produces N_2 completed events in that interval, the primary estimator is

$$\boxed{\widehat{\rho} = \frac{N_2}{M}.}$$

For the ideal system with

$$M = 2,$$

the preregistered records are

$$\boxed{N_2^{(\mathrm{S})} = 6}$$

and

$$\boxed{N_2^{(\mathrm{I})} = 4.}$$

20 Primary experimental outcome

The primary outcome is not the fitted oscillation frequency.

It is the synchronized internal event record.

The primary data for each mode are therefore:

$$M,$$

$$N_2,$$

and

$$\widehat{\rho} = \frac{N_2}{M}.$$

Oscilloscope frequency measurements and Fourier spectra are secondary diagnostics.

They may be reported, but they do not replace the predeclared event comparison.

21 Secondary diagnostic measurements

The following may be recorded for engineering diagnosis:

- oscillation frequencies;
- Fourier spectra;
- mode amplitudes;
- DC offsets;
- waveform distortion;
- supply rails;
- comparator switching times;
- temperature;
- long-term drift.

These quantities may explain a failure.

They may not be used retrospectively to redefine the primary structural prediction.

22 S run

The S run proceeds as follows.

1. Verify static calibration has been completed and locked.
2. Select the S feedback branch.
3. RESET all state integrators.

4. PREPARE $u \propto (1, -1, 0)^T$.
5. Confirm $v \approx 0$.
6. Enter RUN.
7. Ignore the arbitrary startup interval for formal counting.
8. Identify the first completed reference event r_1.
9. Open the formal observation window.
10. Continue through reference event r_{M+1}.
11. Close the formal observation window.
12. Record the number of test events $N_2^{(\mathrm{S})}$.
13. Archive the complete raw traces.

No structural parameter is altered during or after this run.

23 I run

The I run uses the same procedure.

The only intended dynamical change is the upper feedback selection:

$$K_B \longrightarrow H_B.$$

All other elements remain fixed:

- K_A;
- integrator values;
- preparation;
- threshold values;
- readout definitions;
- observation length M;
- event algorithm;
- calibration procedure.

The resulting test-event count is recorded as

$$N_2^{(\mathrm{I})}.$$

24 Reference-channel control

The reference mode has the same ideal rate in both models.

Therefore a consistency check is

$$\boxed{\Omega_{1,\mathrm{S}} \approx \Omega_{1,\mathrm{I}}.}$$

A statistically or engineering-significant reference-channel difference beyond the precomputed circuit uncertainty is treated as an apparatus inconsistency.

The S/I comparison is not accepted as a clean model test until that inconsistency is understood.

25 Predeclared experimental classifications

The experiment may end in one of several scientifically distinct outcomes.

25.1 PASS-S

The observed S configuration agrees with its independently calibrated prediction and is incompatible with the I prediction band.

25.2 PASS-I

The observed I configuration agrees with its independently calibrated prediction and is incompatible with the S prediction band.

Because both circuit modes are intentionally constructed, this terminology refers to correct realization of each conditional dynamics, not discovery of which law nature fundamentally selects.

25.3 DUAL REALIZATION PASS

The intended primary engineering outcome is that both selectable physical circuit configurations reproduce their respective pre-run predictions.

This establishes that the common apparatus can realize both conditional models.

25.4 CALIBRATION FAILURE

The static matrices cannot be measured with sufficient precision to produce separated pre-run prediction intervals.

No dynamical claim is made.

25.5 MODE-PURITY FAILURE

One or more modal readouts violate the predeclared hysteretic mode-purity condition.

The event classification is considered invalid.

25.6 REFERENCE-CONTROL FAILURE

The common reference channel behaves differently between S and I beyond the calibrated expectation.

The result is treated as an apparatus inconsistency.

25.7 MODEL-DISAGREEMENT RESULT

The apparatus passes structural calibration and control checks, but the transient record disagrees with the prediction of the calibrated state matrix.

This is a genuine negative experimental result and must be reported as such.

26 Prohibited post-run operations

The following actions are prohibited after examining the formal transient result:

- changing K_A to improve agreement;
- changing K_B to improve agreement;
- changing H_B to improve agreement;
- redefining integrator gains from observed oscillation frequencies;
- changing readout definitions;
- changing hysteresis thresholds solely because of the observed classification;
- changing M after seeing which count was obtained;
- excluding valid trials because they disagree with prediction.

Any later engineering modification produces a new experimental revision and must be labeled accordingly.

27 Repeatability protocol

A single successful run is not sufficient for the final experimental report.

The completed experiment should include repeated independently reset trials.

Before data acquisition, the final protocol should specify:

$$n_{\mathrm{S}}$$

S trials and

$$n_{\mathrm{I}}$$

I trials.

Each trial begins with a fresh RESET/PREP/RUN sequence.

The complete count distribution must be reported.

No unsuccessful valid trial is discarded.

28 Recommended trial structure

A suitable initial engineering campaign is:

10

S trials and

10

I trials after calibration.

If all twenty trials produce stable and clean records, a larger repeat set may subsequently be run.

The distinction between the initial predeclared experiment and any later replication series should remain explicit.

29 Raw-data preservation

The following raw data should be archived for every formal trial:

- u_1, u_2, u_3;
- v_1, v_2, v_3;
- X_1;
- X_2;
- reference comparator output;
- test comparator output;
- QBAR reference event stream;
- QBAR test event stream;
- MODE state;
- RESET state;
- RUN state;
- supply rails if available;
- oscilloscope time base.

The primary event calculation should be reproducible directly from the archived digital event traces.

30 Calibration-data preservation

The calibration archive should contain:

$$G_u, \qquad G_v,$$

$$K_A, \qquad K_B, \qquad H_B,$$

together with:

- all bipolar injection measurements;
- all integrator ramp measurements;
- threshold-voltage measurements;
- component identifiers;
- board revision;
- calibration date;
- measurement-instrument identifiers.

31 Component and apparatus record

The final manuscript should include a complete BOM and schematic revision.

At minimum, the design presently contains:

- six principal analog integrators;
- two physical K networks;
- one calibrated I-branch identity network;
- analog RESET/HOLD switching;
- S/I mode routing;
- two modal readout networks;
- upper and lower threshold comparators;
- hysteretic event latches;
- digital event counters;
- calibration-access headers;
- analog test points.

The final component manufacturer and lot information should be inserted after construction.

32 Mixed-model SPICE prerequisite

Before physical construction results are interpreted, the frozen engineering program calls for an actual mixed-model SPICE execution.

The intended stack contains:

- the OPAx197-family manufacturer macromodel;
- the ADG333A switching model;
- the ADG1412 switching model;
- a transparent datasheet-bounded behavioral model for the LM319 comparator.

The SPICE simulation must follow the same structural firewall as the physical experiment.

Its results should be incorporated into the final experimental paper before bench measurements are interpreted.

33 SPICE results

Status at protocol freeze: not yet executed.

This section is intentionally reserved.

The final manuscript should report:

- model versions;
- simulator and version;
- any syntax compatibility modifications;
- static reconstructed matrices;
- predicted S/I modal ratios;
- S transient;
- I transient;
- EW-0 event counts;
- stability result;
- mode-purity result;
- deviations from ideal behavior.

No synthetic result should be inserted into this section.

34 Physical experimental results

Status at protocol freeze: no physical measurement has been performed.

This section is intentionally empty until bench data exist.

The final Results section should report the observations without altering the preceding prediction or Methods sections.

Recommended structure:

34.1 Measured static calibration

Insert measured

$$G_u, \quad G_v, \quad K_A, \quad K_B, \quad H_B.$$

34.2 Pre-run predictions

Insert

$$\rho_{\mathrm{S}}^{\mathrm{cal}},$$

$$\rho_{\mathrm{I}}^{\mathrm{cal}},$$

their uncertainties, and selected M.

34.3 S trial results

Report all valid S counts.

34.4 I trial results

Report all valid I counts.

34.5 Reference-channel controls

Report reference consistency.

34.6 Waveform diagnostics

Report measured traces and spectra.

34.7 Protocol deviations

Report every deviation from the frozen procedure.

35 Interpretation rules

Interpretation is predeclared as follows.

If both selectable circuit configurations reproduce their independent calibrated predictions, the experiment supports the physical realizability of the conditional benchmark.

If one or both fail despite successful static calibration and valid readout conditions, the corresponding hardware realization is falsified at that revision.

If static calibration fails, the dynamical test is inconclusive.

If event purity fails, the count result is inconclusive.

If the reference control fails, the S/I comparison is inconclusive until the apparatus discrepancy is explained.

No experimental outcome establishes HSMT as fundamental physics.

36 Relationship to the previous operational paper

The preceding operational-benchmark paper established the mathematical and engineering construction that motivates this experiment.

The present paper does not reopen those derivations.

Instead, it converts the frozen benchmark into a predeclared physical protocol.

The two papers therefore have distinct roles:

Paper I: derive the benchmark

and

Paper II: physically test the benchmark.

This separation prevents the derivation from being modified in response to the eventual experimental outcome.

37 Discussion

The unusual feature of this experiment is that the central discriminator is dimensionless and internally referenced.

A conventional frequency test would require fitting absolute frequencies against an external time standard.

Here the primary measurement is instead

$$\widehat{\rho} = \frac{N_2}{M},$$

where both numerator and denominator are generated internally by the same device.

This does not mean laboratory time is absent from the physical system.

It means uncertainty in the common absolute clock scale is largely irrelevant to the primary model discriminator.

The scalar-copy invariance derived previously strengthens this property: a broad class of positive multiplicative gain errors changes the overall rate while preserving the modal ratio.

Consequently the experiment focuses naturally on nonscalar structural error, cross-mode contamination, drift, and event integrity rather than absolute frequency calibration.

38 Limitations

The experiment has several explicit limitations.

First, it is an emulator experiment.

Second, the candidate dynamics are intentionally engineered into the apparatus rather than discovered as unknown natural laws.

Third, the experiment therefore tests realization and discrimination, not foundational selection.

Fourth, successful operation does not show that physical spacetime is a transportation fiber.

Fifth, successful operation does not establish an HSMT shell hierarchy in nature.

Sixth, the experiment contains no direct test of $N = 4$, Gaussian shell width, cosmology, gravitation, or gauge structure.

Seventh, any later use of this benchmark as a bridge to natural phenomena would require an independently justified mapping from a physical system to the structural operator K.

39 Conclusion

This paper defines the physical experiment for a minimal clock-free HSMT conditional dynamical benchmark.

The benchmark uses one structural operator and two selectable dynamical completions whose ideal modal-rate predictions are

$$\boxed{3 \quad \text{and} \quad \sqrt{3}.}$$

Both are realized in the same six-state analog architecture.

The experiment employs:

- common structural hardware;
- common preparation;
- common readouts;
- common hysteretic event detection;
- independent static calibration;
- precomputed prediction bands;
- a synchronized internal observation window;
- predeclared acceptance and failure criteria.

No physical data have yet been acquired.

The purpose of fixing the protocol now is precisely to prevent the eventual experimental outcome from influencing the prediction or analysis rules after the fact.

The next scientific action is therefore no longer theoretical.

It is to execute the frozen mixed-model SPICE validation and then construct and measure the physical apparatus.

A Ideal prediction table

Quantity	S model	I model
Reference eigenvalue	1	1
Test eigenvalue	3	3
Reference rate	g	g
Test rate	$3g$	$\sqrt{3}\,g$
Rate ratio	3	$\sqrt{3}$
Ideal M	2	2
Ideal test-event count	6	4

B Pre-run experiment record

Before formal transient acquisition, record:

Board revision: _____
Date: _____
Operator: _____
Supply rails: _____
Thresholds: _____
M : _____
$\rho_{\mathrm{S}}^{\mathrm{cal}}$: _____
$\rho_{\mathrm{I}}^{\mathrm{cal}}$: _____
$\delta\rho_{\mathrm{S}}$: _____
$\delta\rho_{\mathrm{I}}$: _____

The following calibration objects must be archived separately:

$$G_u, \qquad G_v, \qquad K_A, \qquad K_B, \qquad H_B.$$

C Formal trial record

For every trial record:

Trial ID: _____
MODE: _____
Preparation valid: _____
Reference control valid: _____
Mode purity valid: _____
M : _____
N_2 : _____
$\widehat{\rho}$: _____
Protocol deviation: _____

D Experimental firewall

The following sequence is mandatory:

$$\begin{aligned}\text{static calibration} &\longrightarrow \text{calibrated matrices} \longrightarrow \text{prediction} \\ &\longrightarrow \text{predeclared } M \longrightarrow \text{transient} \longrightarrow \text{comparison}.\end{aligned}$$

The reverse flow

$$\text{transient} \longrightarrow \text{structural retuning}$$

is prohibited within a single experimental revision.

Protocol status

At the time this manuscript is frozen:

Physical apparatus: not yet measured

Mixed-model SPICE transient: not yet executed

Predictions and analysis rules: predeclared

References

[1] V. Garrett, *Hierarchical Shell Manifold Theory: Graded Geometry, Intergrade Dynamics, and Effective Physics*, HSMT framework manuscript, 2026.

[2] V. Garrett, *From Relational Configuration Geometry to a Clock-Free Experimental Benchmark in Hierarchical Shell Manifold Theory*, HSMT operational-benchmark manuscript, 2026.